Four Signal-Processing Themes

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Key Concepts

1) Equalization problems are concerned with recovering the input to a "system" from measurement of the output and knowledge of the system characteristics.

2) Equalization usually implies correcting distortion in a signal.

3) Filtering involves determining the output of a system given the input.

4) Filtering implies separation - separating signal from noise or one signal characteristic from another.

5) Identification problems involve using "input" and "output" signals to identify a mathematical model relating the two signals.

6) Estimation and detection problems involve extracting a piece of desired information from a signal.

    a) Estimation involves determining the value of a parameter in a mathematical model for the signal.

    b) Detection involves deciding whether a particular characteristic is present in or absent from the measured signal.

7) Different signal processing techniques are applied to these different problem themes.

Transcript

Four Signal-Processing Themes

Introduction

There are several common problem themes in signal processing that transcend specific applications and methods. In this lesson we will consider four problem themes called

- Equalization
- Filtering
- Identification
- Estimation and detection

Slightly different terminology may be used for these basic problems in different applications, for example sometimes equalization is called inversion.

The first three problems can be interpreted in terms of the block diagram depicted in Figure 1. This diagram depicts an input signal being modified by a system to produce an output signal. Each of the three problems is obtained depending on whether the output, system, or input signal is assumed unknown.

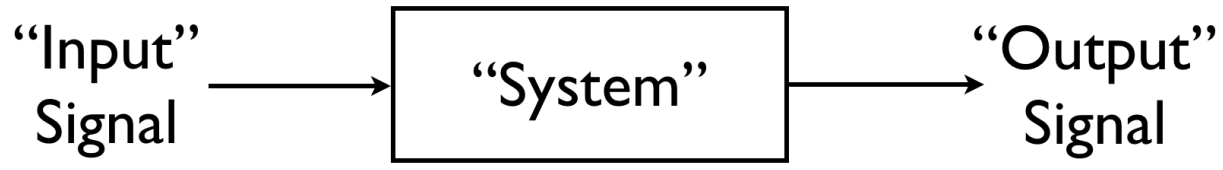


Figure 1. Framework for equalization, filtering, and identification problems.

Equalization

The equalization problem seeks to recover the input signal from knowledge of the system and the output signal. This is also known as “inversion” because it amounts to undoing or inverting the effect of the system on the output signal.

The equalization problem views the output signal as a corrupted or distorted version of the input. The system introduces the distortion.

A famous example of equalization is associated with the initial launch of the Hubble Space Telescope. The images after launch were very blurry because the mirrors had been improperly ground. Signal processing of the blurry images was used to equalize or compensate for the distortion introduced by the incorrectly shaped mirrors until a space shuttle mission repaired the mirrors.

Filtering

The filtering problem involves computing an output signal given the input signal and system characteristics.

In common language the term filtering implies separation. For example, a coffee filter separates liquid coffee from coffee grounds. Filters are often used in signal processing to separate components of the input signal in order to reduce noise and/or interference.

There are two components to the filtering problem: designing a system or set of computations that will apply the desired characteristics to the input, and then implementing those computation on the input signal in order to obtain an output signal.

The filter may be designed to separate signal from noise or to separate one aspect of the input signal from other aspects. Filters may also be designed to shape the characteristics of the input signal in a desired fashion. For example, a filter may be designed to solve the equalization problem.

Shaping the characteristics of the input signal in a desired fashion is sometimes accomplished by first separating the signal into components associated with different frequency ranges, independently adjusting those components, and then recombining the adjusted components.

An example is found in the “equalizers” available in many audio players. Equalizers allow the user to adjust the relative amplitude of different audio frequency components to the user’s preference. This is accomplished using filters to separate the signal into different frequency bands. The user chooses the gain applied to each band. The use of the term equalizer is rooted in their historical role to compensate for distortions introduced by the listening environment.

Identification

The identification problem involves using knowledge of the input and output signals to "identify" the properties of an unknown system or a mathematical model for the action of the system on the input. In the identification problem the "system" normally represents a physical device or process that acts on the input signal.

Identification of the system may be required for subsequent signal processing. For example, signals transmitted in wireless communication applications typically are distorted due to multipath propagation and scattering from objects such as buildings. The system in this case is a mathematical model for the distortion. This model is identified using periodic training inputs and is then used to improve the communication system performance.

In other applications the characteristics of the unknown system are of direct interest. For example, geologists interested in the structure of the earth use explosives to apply an "input" signal to the earth. Seismometers located some distance away measure the resulting vibrations as "output" signals. A mathematical model for the effect of the earth's structure on the input signal is identified and provides useful information for oil and gas prospecting.

Estimation and Detection

Estimation and detection problems involve directly extracting information from a signal. The information is usually contained in one or more parameters that are involved in a model for the signal. One of the hallmarks of estimation and detection methodologies is reliance on statistical models for describing uncertainty in the data.

For example, in a Doppler radar system the relative velocity of an object, such as a speeding automobile, is determined from the Doppler shift of the backscattered signal. The signal-processing goal is to estimate the frequency of the backscattered signal consisting of the signal of interest in noise. The Doppler shift is determined by comparing the transmitted frequency to the estimated frequency of the received signal.

As a second example, in network modeling the goal is to estimate the strength of the network connections from the measured data.

Detection problems are very closely related to estimation and involve detecting or deciding whether a particular feature is present. For example, we may wish to decide whether a particular connection is present or absent in a network model. The name detection originated in the early radar community because they were interested in "detecting" the presence of an enemy aircraft in backscattered radar signals.

Summary

The four problems described in this lesson arise in many different signal processing application areas. Each problem has its own characteristics and toolset. Learning the tools associated with each class of problems and developing and ability to apply those tools is essential for the signal-processing practitioner. The high-level view presented in this lesson prepares you to recognize these categories of problems, the first step in choosing the appropriate tools for their solution.

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